

2021 Mathematics Standard Year 11 – Examination Worked Solutions

1. C

All four options are quantitative data as they are all recorded as numbers.

In options A, B and D, the scores are whole numbers, which is discrete data.

Only option C is continuous as measurements, such as time, are always continuous data.

So time is an example of quantitative, continuous data.

Hence, the correct option is C.

2. A

This line has a positive gradient of $m = \frac{4}{2} = 2$

This line cuts the y axis at 4.

Using the formula $y = mx + b$, the equation must be $y = 2x + 4$.

Hence, the correct option is A.

3. B

$$\$5\,500 - \$450 = \$5\,050$$

If 60% of the remainder was spent on the fridge, 40% must have been spent on the TV.

$$\frac{40}{100} \times \$5\,050 = \$2\,020$$

The TV costs \$2 020

Hence, the correct option is B.

4. D

As Mexico city has a UTC of -6 and Sydney has a UTC of +10, then Mexico city is 16 hours behind Sydney.

The time in Sydney is 2am Sunday. 2 hours before this, the time was midnight Saturday and 14 hours before midnight Saturday, the time was 10am Saturday.

Hence, the correct option is D.

5. A

Jacob's earnings in 2 weeks is $\$650 \times 2 = \1300 .

The extra amount Jacob earned is

$$\$3100 - \$1300 = \$1800.$$

Every 2 items is worth \$75, $\$1800 \div \$75 = 24$.

The number of items sold is $24 \times 2 = 48$.

Hence, the correct option is A.

6. B

The total number of points scored in the first 5 games is $180 \times 5 = 900$.

The total number of points scored in the first 8 games is $8 \times 183 = 1464$.

The total points scored in the last 3 games is $1464 - 900 = 564$.

The average per game for the last 3 games is $564 \div 3 = 188$.

Hence, the correct option is B.

7. C

The formula for the volume of a cylinder is :-

$$V = \pi r^2 h$$

So $\pi r^2 h = 1392$

$$\pi r^2 \times 10 = 1392$$

$$r^2 = \frac{1392}{10\pi}$$

$$r = \sqrt{\frac{1392}{10\pi}}$$

$$r = 6.656...$$

The radius is 6.7 cm correct to 1 decimal place.

Hence, the correct option is C.

8. D

The absolute error is half of the smallest unit measured.

The absolute error is $\frac{1}{2} \times 1\text{cm} = 0.5\text{cm}$

Therefore, the percentage error is

$$\pm \frac{0.5}{200} \times 100 = \pm 0.25\%$$

Hence, the correct option is D.

9. C

Substituting what we are given into the equation :-

$$3 = \frac{\text{age} \times 25}{150}$$

Let x be the child's age

$$3 = \frac{25x}{150}$$

$$25x = 450$$

$$x = \frac{450}{25}$$

$$x = 18$$

Therefore, this child is 18 months old.

Hence, the correct option is C.

10. A

For his first selection there are 27 lollies and 16 are honey flavoured, the probability of selecting a honey flavoured lolly is $\frac{16}{27}$.

For his second selection there are 26 lollies and if the first lolly selected is honey flavoured, just 15 are honey flavoured, the probability of selecting a honey flavoured lolly is $\frac{15}{26}$.

So the probability that both of the lollies selected by Andrew are honey flavoured is found by multiplying these 2 probabilities $\frac{16}{27} \times \frac{15}{26}$.

Hence, the correct option is A.

11. B

Looking at the arc for the cumulative percentage, the dot for "Over Priced" is closest to 69%.

The previous category, "Voice Quality" had a cumulative percentage of approximately 53%.

By subtraction, $69\% - 53\% = 16\%$, this is the amount for those who chose "Over Priced."

Alternative Method:

The number of customers unhappy because of the overpriced is 192.

The total number of customers is

$$384 + 264 + 192 + 168 + 120 + 72 = 1200$$

Therefore, the percentage of customers unhappy because of the overpriced is

$$\frac{192}{1200} \times 100 = 16\%$$

Hence, the correct option is B.

12. D

David has already saved \$400, so he still needs

$$\$2\,800 - \$400 = \$2\,400$$

Each week he saves \$120, so in t weeks he saves $120t$ dollars. This indicates he still needs

$$2400 - 120t \text{ dollars}$$

$$\text{Hence, } M = 2\,400 - 120t$$

Hence, the correct option is D.

13. A

	Run		
			
Shapes (S)	1	2	3
Matchsticks (N)	8	14	20
			
	Rise		

$$\text{Rise} = 14 - 8 = 6$$

$$\text{Run} = 2 - 1 = 1$$

$$\text{Therefore, gradient} = \frac{6}{1} = 6$$

$$6 \times 1 + 2 = 8$$

$$6 \times 2 + 2 = 14 \text{ and so on.}$$

$$\text{Then, } N = 6S + 2$$

$$6S + 2 = 200$$

$$6S = 198$$

$$\text{So, } S = 33$$

Alternative Method:

The first shape uses 8 matches, each extra shape requires 6 matches. So N , the number of matches needed for S shapes could be found using the formula

$$N = 6S + 2$$

$$6S + 2 = 200$$

$$6S = 198$$

$$\text{So, } S = 33$$

So the maximum number of shapes which could be made with 200 matches is 33.

Hence, the correct option is A.

14. C

When we multiply each score by 1.2, the original mean and standard deviation are multiplied by 1.2.

$$\text{So, the original mean is } 66 \div 1.2 = 55 \text{ and}$$

$$\text{the original standard deviation is } 10.2 \div 1.2 = 8.5.$$

Hence, the correct option is C.

15. B

The area of a quarter of a circle with a radius of

10cm would be $A = \frac{1}{4} \times \pi \times 10^2$

$$A \approx 78.5398 \dots \text{cm}^2$$

The area of half of a circle with a diameter of 10cm

($r = 5$) would be $A = \frac{1}{2} \times \pi \times 5^2$

$$A \approx 39.2699 \dots \text{cm}^2$$

The shaded area must be $78.5398 - 39.2699$

$$= 39.2699 \text{ cm}^2$$

This would be 39.3 cm^2 when written with

3 significant figures.

Hence, the correct option is B.

Section II**QUESTION 16**

$$\text{a) } \bar{x} = \frac{350+470+530+540+380+480+460}{7}$$

$$\bar{x} = \frac{3210}{7}$$

$$\bar{x} = \$458.57$$

b) Entering the data into a calculator, the standard deviation is 65.78 to two decimal places.

QUESTION 17

As the job is more than 2 hours, the call out fee will be discounted by 60%, which means the customer only pays 40 % of \$ 90.

Call Out Fee:

$$\frac{40}{100} \times \$90 = \$36$$

3 Hours Cost:

$$3 \times \$80 = \$240$$

$$\text{The total cost will be } \$36 + \$240 = \$276$$

QUESTION 18

Note, 1 g fat = 37 kilojoules

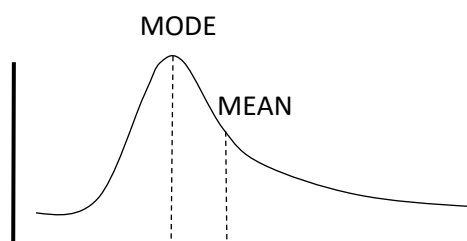
Therefore, 12.2 g fat = $12.2 \times 37 = 451.4$ kilojoules

Also note, 4.184 kilojoules = 1 kilocalorie

$$1 \text{ kilojoule} = \frac{1}{4.184} \text{ kilocalories}$$

$$\text{Hence, } 451.4 \text{ kilojoules} = \frac{451.4}{4.184} \text{ kilocalories}$$

$$= 107.89 \text{ kilocalories}$$

QUESTION 19**QUESTION 20**

$$2400 \text{ W} = 2.4 \text{ kW}$$

$$28 \text{ hours at } 2.4 \text{ kW} = 67.2 \text{ kWh}$$

$$\text{Cost for 4 weeks} = 67.2 \times \$0.25 = \$16.80$$

$$\text{Cost per week} = \$16.80 \div 4 = \$4.20$$

QUESTION 21

Mario will be paid \$80 for the first 2 hours

He will earn time and a half for the final 6 hours

$$\$24 \times 1.5 \times 6 = \$216$$

$$\text{Mario's total earnings will be } \$80 + \$216 = \$296$$

QUESTION 22

$$\text{Samantha's deposit is } \frac{20}{100} \times \$15\,000 = \$3\,000$$

$$\text{The amount borrowed is } \$15\,000 - \$3\,000 = \$12\,000$$

$$\text{The amount repaid} = \$717 \times 18 = \$12\,906$$

$$\text{The interest charged is } \$12\,906 - \$12\,000 = \$906$$

Using the simple interest formula, $I = PRN$

$$906 = 12\,000 \times r \times 1.5$$

$$18\,000 \times r = 906$$

$$r = 0.050333 \dots$$

The simple interest charged is 5.03 %

QUESTION 23

Joanne has a monthly income of \$4 000 and

average expenses of \$2480, meaning she can save

$$\$4\,000 - \$2\,480 = \$1\,520 \text{ per month.}$$

In 6 months Joanne will be able to save

$$6 \times \$1\,520 = \$9\,120$$

QUESTION 24

Let the cost of the ring before GST be \$ x . The GST paid on it will be $0.12x$. So, the total money paid is

$$x + 0.12x = 2\,408$$

$$1.12x = 2\,408$$

$$x = 2\,408 \div 1.12$$

$$x = \$2\,150$$

Hence, the original price was \$ 2150 and the total paid is \$ 2408 so the GST is

$$\$2\,408 - \$2\,150 = \$258$$

QUESTION 25

The interest rate is 18% p.a. = $\frac{0.18}{365}$ per day.

Using the compound interest formula,

$$FV = 2800 \left(1 + \frac{0.18}{365}\right)^{15}$$

$$FV = \$2\,820.78$$

Raymond still owes

$$\$2\,820.78 - \$1\,150 = \$1\,670.78$$

QUESTION 26

$$a) Q_1 = \frac{29+29}{2} = 29$$

$$Q_2 = \frac{36+38}{2} = 37$$

$$IQR = 37 - 29 = 8$$

b) Using $Q_1 - 1.5 \times IQR$, the lower boundary is $29 - 1.5 \times 8 = 29 - 12 = 17$.

As 15 is less than 17, this means 15 does not fit in the boundary, so it is an outlier.

QUESTION 27

The time taken to receive 180mL is 1 hour and 40 minutes which is 100 minutes.

Since the medicine was delivered at a constant rate, this means the patient received

$$\frac{180}{100} = 1.8 \text{ mL of medicine per minute.}$$

Using ratio,

$$\text{mL} \quad \text{min} \quad \text{mL} \quad \text{min}$$

$$1.8 : 1 = 225 : x$$

$$x = \frac{225}{1.8} \times 1 = 125 \text{ minutes.}$$

Hence, the patient received a dose of 225 mL in 125 minutes which is 2 hours and 5 minutes.

QUESTION 28

$$a) V_{\text{cylinder}} = \pi \times 8^2 \times 18 \\ = 3619.114 \dots \text{ cm}^3$$

$$V_{\text{cone}} = \frac{1}{3} \times \pi \times 8^2 \times 6 \\ = 402.123 \dots \text{ cm}^3$$

$$\therefore V_{\text{solid}} = 3619.114 \dots - 402.123 \dots \\ \approx 3216.99 \text{ cm}^3$$

$$b) \text{ The curved area of the cylinder} = 2\pi rh \\ = 2 \times \pi \times 8 \times 18 \\ = 904.778 \dots \text{ cm}^2$$

$$\text{Bottom area} = \pi \times 8^2 = 201.061 \dots \text{ cm}^2$$

Hence total surface area

$$= 80\pi + 904.778 \dots + 201.061 \dots \\ \approx 1357.17 \text{ cm}^2$$

QUESTION 29

a) Pamela's tax deductions are

gross income – taxable income

$$\text{which is } 105\,260 - 101\,936 = \$3\,324$$

As a percentage of gross income

$$= \frac{3\,324}{105\,260} \times 100 \\ = 3.16 \%$$

b) Pamela's tax on her taxable income is

$$(101\,936 - 87\,000) \times 0.37 + 19\,822 \\ = \$25\,348.32$$

$$c) \text{ Medicare levy} = \frac{2}{100} \times \$101\,936 \\ = \$2\,038.72$$

d) Medicare levy + tax

$$= \$2\,038.72 + \$25\,348.32 \\ = \$27\,387.04$$

As Pamela received a refund of \$ 2 140, she must have paid this much more than required.

Her PAYG tax must have been

$$\$27\,387.04 + \$2\,140 = \$29\,527.04.$$

QUESTION 30

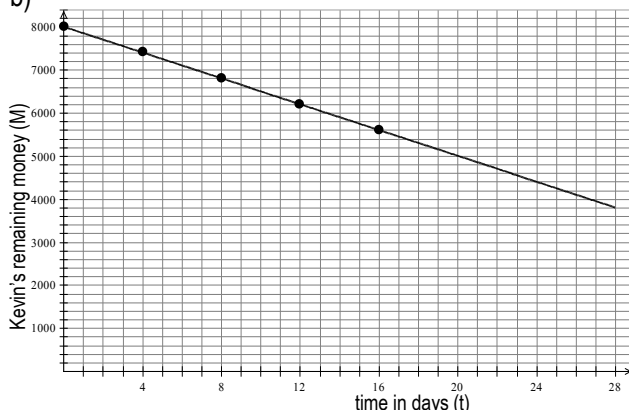
a) $\$7\,400 - \$600 = \$6\,800$

$\$6\,200 - \$600 = \$5\,600$

The completed table is :-

t	0	4	8	12	16
M	8000	7400	6 800	6200	5 600

b)



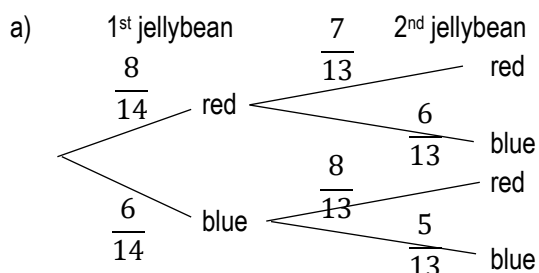
c) Brett's money is decreasing at a rate of \$600 every 4 days, this is mathematically equivalent to \$150 per day, so after t days he has reduced his money by $\$150t$.
Initially, when $t = 0$, he has \$8 000.

So the amount of money Brett has remaining after t days can be expressed as $M = 8\,000 - 150t$.

d) Solve $2\,600 = 8\,000 - 150t$
 $8\,000 - 150t = 2\,600$
 $-150t = -5\,400$
 $t = \frac{-5\,400}{-150}$
 $t = 36$

Hence, Brett will have \$2 600 remaining after 36 days.

QUESTION 31



b) Three of these outcomes contain at least 1 red jellybean.

$$P(RR) = \frac{8}{14} \times \frac{7}{13} = \frac{4}{13}$$

$$P(RB) = \frac{8}{14} \times \frac{6}{13} = \frac{24}{91}$$

$$P(BR) = \frac{6}{14} \times \frac{8}{13} = \frac{24}{91}$$

$$\therefore P(\text{at least 1 red}) = \frac{4}{13} + \frac{24}{91} + \frac{24}{91} = \frac{76}{91}$$

Alternative method:

Using complementary events, we get:

$$P(\text{at least 1 red}) = 1 - P(\text{no red}) = 1 - \frac{6}{14} \times \frac{5}{13} = \frac{76}{91}$$

QUESTION 32

Kevin's energy intake:

$$2 \times 150 + 170 = 470 \text{ kilocalories}$$

As 1 kilocalorie = 4.184 kJ, so Kevin's intake is
 $470 \times 4.184 \text{ kJ} = 1966.48 \text{ kJ}$.

Kevin's energy used:

At a speed of 12km/h, he will use 10.54 kJ per kg for every 30 minutes and as his mass is 84kg, then he will use $10.54 \times 84 = 885.36 \text{ kJ}$ per 30 minutes.

The time he needs to ride his bike to burn off the energy intake is

$$\frac{1966.48}{885.36} = 2.221107798 \text{ lots of 30 minutes.}$$

Hence, the time needed is

$$30 \times 2.2211... = 66.63... \text{ min} \approx 67 \text{ minutes}$$

QUESTION 33

The amount of water added by the storm would be
 $3\,959 - 200 = 3\,759 \text{ litres}$.

As with each minute, 35.8 litres of water are added to the tank by the storm, then the time that the storm lasted is
 $3759 \div 35.8 = 105 \text{ minutes} = 1 \text{ hour } 45 \text{ minutes}$

Hence, the storm lasted 1 hour 45 minutes.

QUESTION 34

Boys:

$$Q_1 = \frac{170+174}{2} = 172 \text{ and } Q_3 = \frac{186+186}{2} = 186$$

$$\text{IQR} = 186 - 172 = 14$$

Girls:

$$Q_1 = \frac{160+160}{2} = 160 \text{ and } Q_3 = \frac{170+170}{2} = 170$$

$$\text{IQR} = 170 - 160 = 10$$

The difference in the interquartile ranges of the boys' group and girls' group is $14 - 10 = 4$, this means the difference in the ranges of the two groups is also 4.

As the range of the boys' group is $190 - 170 = 20$, so the range of the girls' group must be 4 more than 20, which is 24.

To find the highest score, which is the missing score, we add 24 with the lowest score.

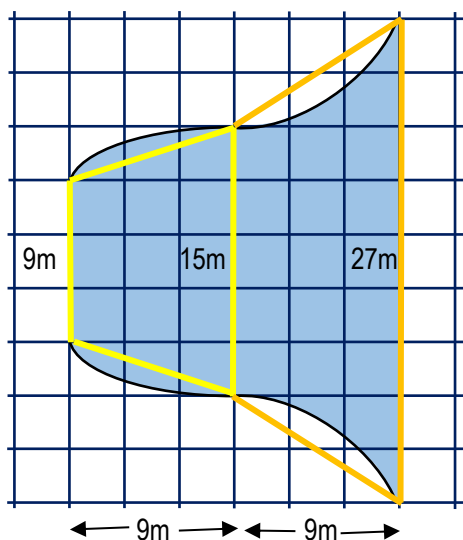
This means the missing score is $153 + 24 = 177$.

Hence, the missing digit in Rebecca's height is 7.

QUESTION 35

a)

Scale 1 cm = 3 m



1st application trapezoidal rule:

$$A(\text{trapezium in yellow}) \approx \frac{9}{2} (9 + 15) \approx 108 \text{ m}^2$$

2nd application trapezoidal rule:

$$A(\text{trapezium in orange}) \approx \frac{9}{2} (15 + 27) \approx 189 \text{ m}^2$$

Hence the approximate area of the pool is

$$108 + 189 = 297 \text{ m}^2.$$

b) The yellow trapezium in the 1st application gives an area which is a bit smaller than the actual area.

The orange trapezium in the 2nd application gives an area which is a bit larger than the actual area.

The extra gained in the 2nd application (orange trapezium) is more than the area lost in the 1st application (yellow trapezium).

So, overall, the area obtained by the two applications is slightly larger than the actual area of the pool. This indicates that Clara's statement is not accurate.

QUESTION 36

At 1am, Steven's BAC was zero. He stopped drinking 50 minutes before that, that is 12:10am.

At that time, his BAC was

$$\frac{50}{60} = \frac{\text{BAC}}{0.015}, \text{ that is his BAC was } 0.0125.$$

Steven's mass $M = 80$ kg and the number of drinks he consumed is $N = 1.24 \times 2 = 2.48$ standard drinks.

By substituting the BAC, M and N into the BAC_{male} formula, we can calculate H , the number of hours he was drinking.

$$0.0125 = \frac{10 \times 2.48 - 7.5H}{6.8 \times 80}$$

$$0.0125 = \frac{24.8 - 7.5H}{544}$$

$$24.8 - 7.5H = 6.8$$

$$-7.5H = -18$$

$$H = 2.4$$

This shows that Steven was consuming alcohol for 2.4 hours or 144 minutes.

He started to drink p minutes after arriving and stopped drinking after $4p$ minutes. This indicates he was drinking for $4p - p = 3p$ minutes.

$$\text{Hence, } 3p = 144$$

$$p = 48 \text{ minutes}$$

The time from his arrival at the party until he stopped drinking at 12:10am is $4p$ which is $48 \times 4 = 192$ minutes = 3 hours 12 minutes.

Hence, Steven arrived at the party 3 hours 12 minutes before 12:10 am which is 8:58 pm.